

AI WEATHERWISE API PROJECT DOCUMENTATION.

Introduction

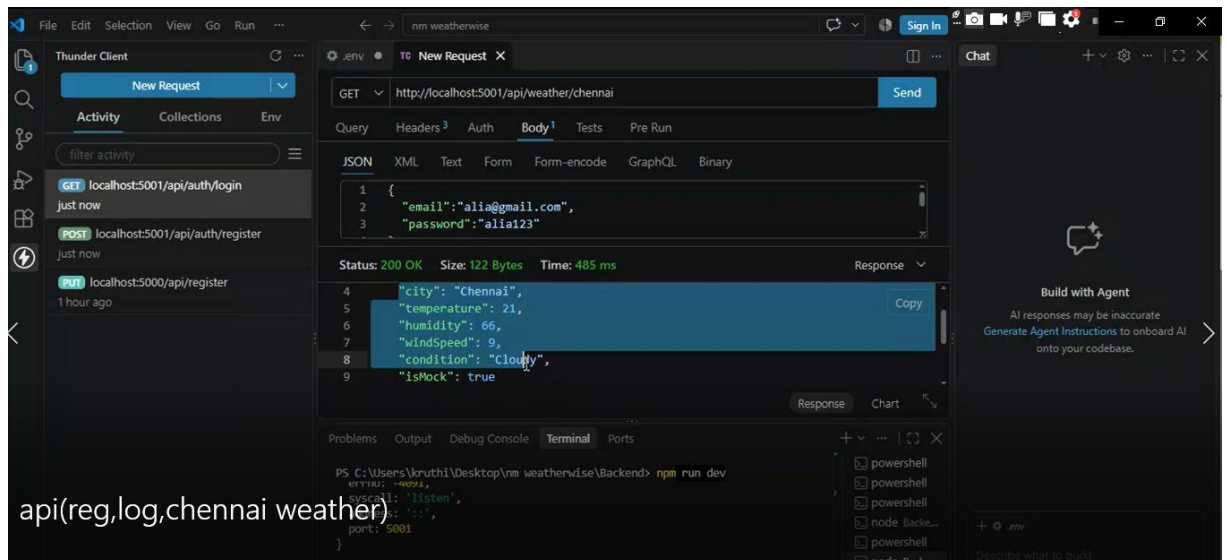
Project Title: "AI Weatherwise Api".

Team Members:

- 1.Gopika.K
- 2.Mathivadhani.T
- 3.Monica.K
- 4.Abarna Balaji.R

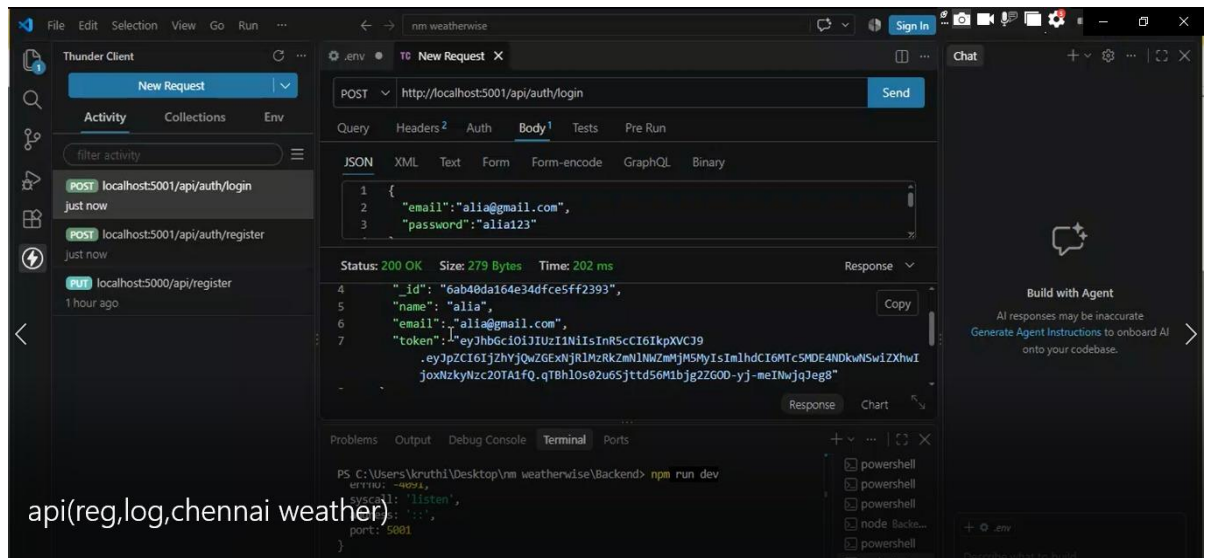
1. Weather API Request (/api/weather/chennai)

- **Function:** Demonstrates fetching current weather data using a GET request to <http://localhost:5001/api/weather/chennai>.
- **Response:** Returns a 200 OK status with details such as city: "Chennai", temperature: 21, humidity: 66, windSpeed: 9, and condition: "Cloudy".



2. User Login Request (/api/auth/login)

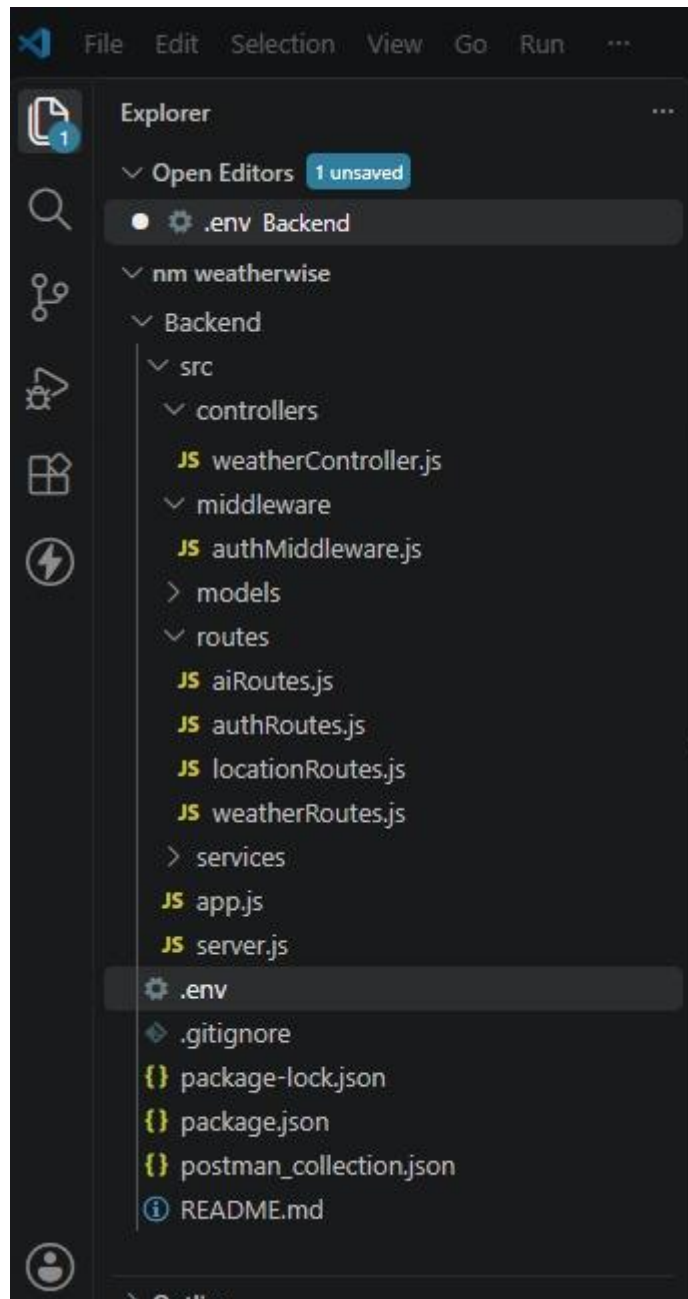
- **Function:** Demonstrates authenticating an existing user via a POST request to `http://localhost:5001/api/auth/login`.
- **Request Body:** Sends user credentials containing email ("alia@gmail.com") and password ("alia123").
- **Response:** Returns a 200 OK status along with user profile information (`_id`, `name`, `email`) and a JSON Web Token (`token`) for session authorization.



3&4.User Registration Request (/api/auth/register)

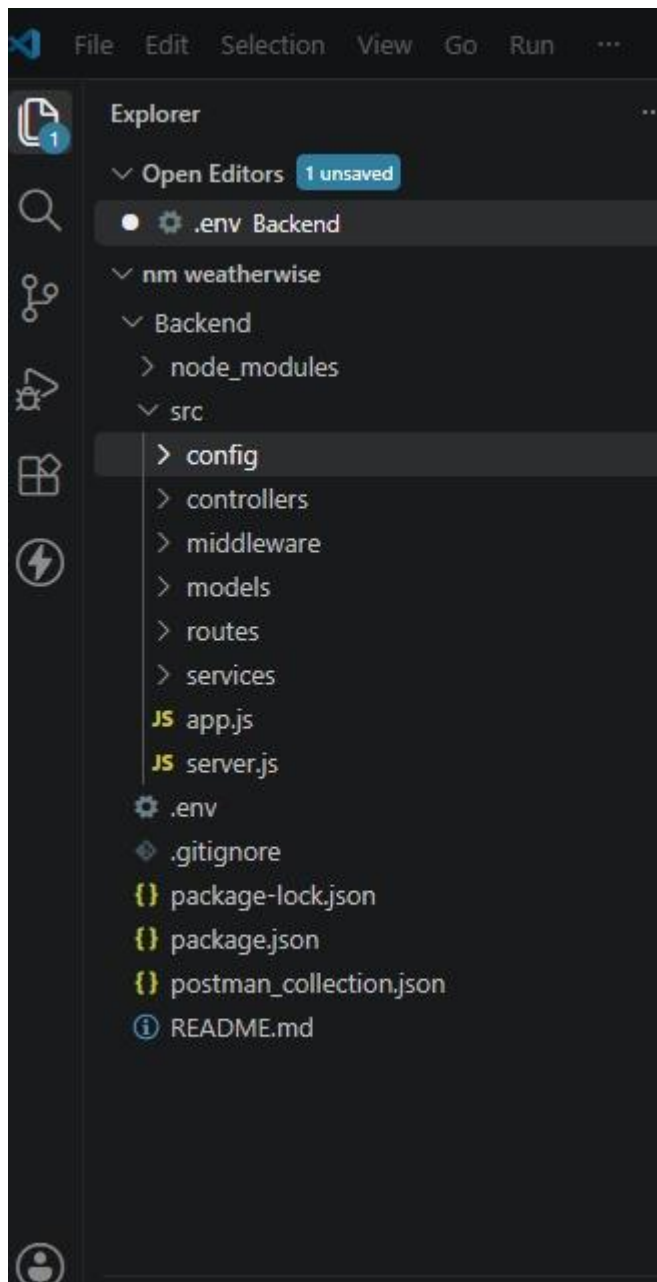
- **Function:** Demonstrates registering a new user via a POST request to `http://localhost:5001/api/auth/register`.
- **Request Body:** Transmits account registration fields such as name and email.
- **Response:** Returns a 201 Created status confirming success: true and outputs the created user object inside data.

- routes/: Express route handlers like aiRoutes.js, authRoutes.js, locationRoutes.js, and weatherRoutes.js.
- Core backend files: app.js, server.js, and .env.



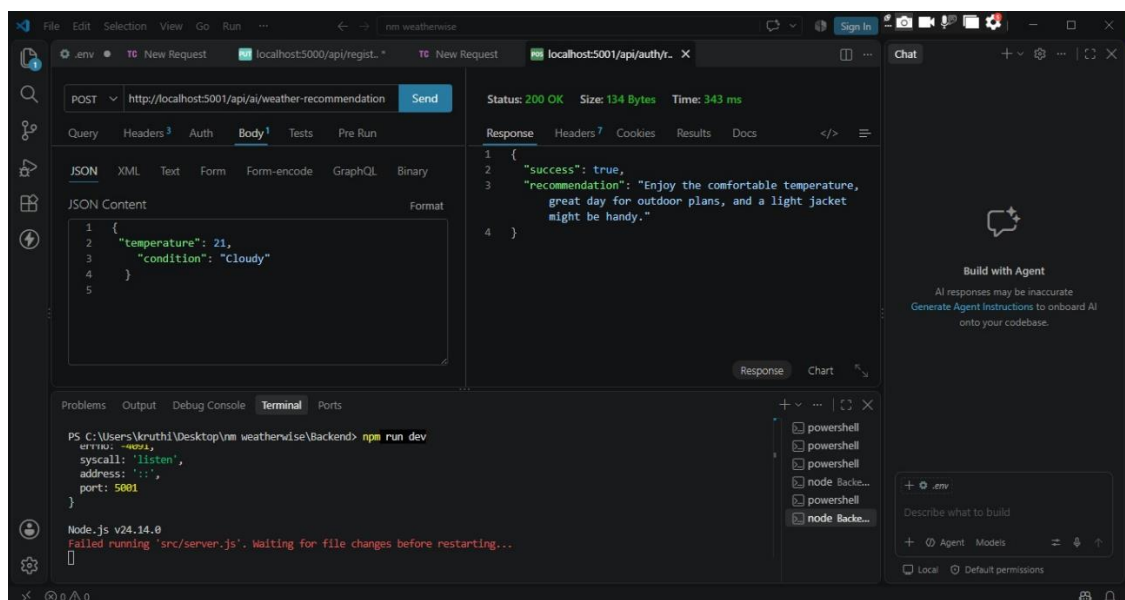
6. Backend Directory Overview (Collapsed)

- **Function:** Displays the high-level architecture of the Backend/src directory.
- **Structure:** Highlights core system folders including config, controllers, middleware, models, routes, and services alongside configuration files like package.json and postman_collection.json.



7.AI Weather Recommendation Request (/api/ai/weather-recommendation)

- **Function:** Demonstrates fetching an automated recommendation based on current conditions using a POST request to `http://localhost:5001/api/ai/weather-recommendation`.
- **Request Body:** Sends temperature and condition data (temperature: 21, condition: "Cloudy").
- **Response:** Returns a 200 OK status containing an AI-generated suggestion: *"Enjoy the comfortable temperature, great day for outdoor plans, and a light jacket might be handy."*



8. AI Weather Summary Request (/api/ai/weather-summary)

- **Function:** Tests generating an automated text summary of weather conditions using a POST request to `http://localhost:5001/api/ai/weather-summary`.
- **Request Body:** Transmits location and weather metrics (city: "Chennai", temperature: 21).
- **Response:** Returns a 200 OK status with a formatted summary message: *"Today's weather in Chennai is pleasant and cloudy with moderate humidity."*

